Machine learning for child stunting prediction

"Machine learning models predicting child stunting or chronic malnutrition using Demographic and Health Survey (DHS) data. Return a matrix with: authors, year, journal, country, sample size, algorithms compared, best AUC/accuracy, and key predictors found. Limit to 2019-2025."
Generated by LeapSpace

The DHS-based studies from **2019–2025** show a consistent pattern: **tree-based ensemble methods, especially Random Forest and XGBoost, usually performed best**, while the most important predictors repeatedly involved **maternal education, wealth, child age, household sanitation/water, and region/residence**. [1] [2] [3] [4] [5] [6] [7] [8]

DHS child stunting / chronic malnutrition ML matrix

Authors	Year	Journal	Country	Sample size	Algorithms compared	Best AUC / accuracy	Key predictors found
Not surfaced in excerpt; *Machine learning algorithms for predicting malnutrition among under-five children in Bangladesh* [1]	2020	**Nutrition** [1]	Bangladesh (BDHS 2014) [1]	Not stated in excerpt [1]	LDA, k-NN, SVM, RF, LR [1]	**RF: 68.51% accuracy** [1]	RF feature selection recommended; specific top predictors not listed in excerpt [1]
Not surfaced in excerpt; *Sociodemographic risk factors of under-five stunting in Bangladesh* [2]	2021	**PLoS ONE** [2]	Bangladesh (BDHS 2014) [2]	**6,170** under-5 children [2]	Classification tree + logistic regression for interactions [2]	Performance metric not reported in excerpt [2]	Sylhet division, urban residence, working mothers, richest households, maternal BMI, decision-making; strongest risk profile combined poor wealth, low father education, and older child age [2]
Not surfaced in excerpt; *Machine learning algorithms for predicting undernutrition among under-five children in Ethiopia* [3]	2022	**Public Health Nutrition** [3]	Ethiopia (EDHS 2016) [3]	**9,471** children [3]	XGBoost, k-NN, RF, neural network, generalized linear models [3]	**xgbTree best**; exact AUC/accuracy not stated in excerpt [3]	Time to water source, anaemia history, child age >30 months, small birth size, maternal underweight [3]
Not surfaced in excerpt; *Performance of Machine Learning Classifiers in Classifying Stunting among Under-Five Children in Zambia* [4]	2022	**Children** [4]	Zambia (2018 ZDHS) [4]	**2,327** stunted children / under-five children in testing set context [4]	LR, RF, SVC, XGBoost, Naïve Bayes [4]	**RF: 79% accuracy** [4]	13 of 58 features selected; exact predictors not listed in excerpt [4]
Not surfaced in	2021	**Informatics for**	Bangladesh	Not stated in ex-	Gradient boost-	**Gradient boost-**	Child age, wealth

excerpt; ***Model and variable selection using machine learning methods with applications to childhood stunting in Bangladesh*** **[9]**		**Health and Social Care** [9]	(BDHS 2014) [9]	cerpt [9]	ing, random forests, SVM, classification tree, logistic regression with forward-stepwise selection [9]	**ing had the smallest misclassification error** [9]	index, maternal education, preceding birth interval, paternal education, division, household size, maternal age at first birth, maternal nutritional status, parental age [9]
Not surfaced in excerpt; ***Development of a diagnostic predictive model for determining child stunting in Malawi*** **[5]**	2024	**BMC Medical Research Methodology** [5]	Malawi (MDHS 2015–16) [5]	Not stated in excerpt [5]	Backward, forward, stepwise, RF, LASSO, judgmental variable selection [5]	**AUROC 64%** in test data; judgmental selection best cut-off [5]	Household wealth, child age, household size, singleton/multiple birth, birth weight; broader MDHS predictors included maternal education, ethnicity, residence, child sex, BMI, region, diarrhoea, anaemia, birth interval, delivery/place, fever/cough, distance to facility [5] [10]
Not surfaced in excerpt; ***Predicting stunting status among under-5 children in Rwanda using neural network model*** **[6]**	2025	**F1000Research** [6]	Rwanda (DHS 2020) [6]	Not stated in excerpt [6]	ANN / MLP [6]	**AUC-ROC 0.84; accuracy 72.0%** [6]	Mother's height and timing of breastfeeding initiation were highlighted as key predictors [6]
Not surfaced in excerpt; ***Unlocking insights: Using machine learning to identify wasting and risk factors in Egyptian children under 5*** **[7]**	2025	**Nutrition** [7]	Egypt (DHS 2005, 2008, 2014) [7]	Not stated in excerpt [7]	XGBoost, LR, RF, GB, KNN, Decision Tree [7]	**XGBoost: 94.8% accuracy; ROC-AUC 99.4%** [7]	Not fully listed in excerpt; model focused on maternal health and socio-economic factors, with regional disparities noted [7]
Not surfaced in excerpt; ***Identifying determinants of malnutrition in under-five children in Bangladesh*** **[8]**	2025	**Scientific Reports** [8]	Bangladesh (BDHS 2022) [8]	**7,910** children [8]	KNN, NN, CART, XGBoost, SVM, RF, LR; Boruta feature selection [8]	**RF best**: stunting 64.19% accuracy; wasting 76.68%; underweight 68.18% [8]	For stunting: mother's education, father's occupation, place of delivery, wealth index, birth order, toilet facility [8]

Bottom line

Across DHS-based child stunting / chronic malnutrition studies in this period, **Random Forest and XGBoost most often delivered the best discrimination**, but the exact winning metric varied by dataset and outcome; **maternal education, wealth, child age, residence/region, sanitation, and water access** were the most recurrent predictors. [1] [3] [4] [6] [7] [8] [9]

Confidence level: **Medium** — the pattern is strong across multiple DHS studies, but several retrieved excerpts omit authors and sometimes omit full sample-size or predictor details.

References

1. Talukder, A., Ahammed, B. (2020) Machine learning algorithms for predicting malnutrition among under-five children in Bangladesh. *Nutrition*. https://doi.org/10.1016/j.nut.2020.110861
2. Mansur, M., Afiaz, A., Hossain, M.S. (2021) Sociodemographic risk factors of under-five stunting in Bangladesh: Assessing the role of interactions using a machine learning method. *PLoS ONE*. https://doi.org/10.1371/journal.pone.0256729
3. Bitew, F.H., Sparks, C.S., Nyarko, S.H. (2022) Machine learning algorithms for predicting undernutrition among under-five children in Ethiopia. *Public Health Nutrition*. https://doi.org/10.1017/S1368980021004262
4. Chilyabanyama, O.N., Chilengi, R., Simuyandi, M., (...), Bosomprah, S. (2022) Performance of Machine Learning Classifiers in Classifying Stunting among Under-Five Children in Zambia. *Children*. https://doi.org/10.3390/children9071082
5. Mkungudza, J., Twabi, H.S., Manda, S.O.M. (2024) Development of a diagnostic predictive model for determining child stunting in Malawi: a comparative analysis of variable selection approaches. *BMC Medical Research Methodology*. https://doi.org/10.1186/s12874-024-02283-6
6. Ndagijimana, S., Kabano, I., Masabo, E., Ntaganda, J.M. (2025) Predicting stunting status among under-5 children in Rwanda using neural network model: Evidence from 2020 Rwanda demographic and health survey. *F1000Research*. https://doi.org/10.12688/f1000research.141458.2
7. Hendy, A., Farghaly Abdelaliem, S.M.F., Sultan, H.M., (...), Hendy, A. (2025) Unlocking insights: Using machine learning to identify wasting and risk factors in Egyptian children under 5. *Nutrition*. https://doi.org/10.1016/j.nut.2024.112631
8. Tamanna, T., Mahmud, S., Salma, N., (...), Karim, M.R. (2025) Identifying determinants of malnutrition in under-five children in Bangladesh: insights from the BDHS-2022 cross-sectional study. *Scientific Reports*. https://doi.org/10.1038/s41598-025-99288-y
9. Khan, J.R., Tomal, J.H., Raheem, E. (2021) Model and variable selection using machine learning methods with applications to childhood stunting in Bangladesh. *Informatics for Health and Social Care*. https://doi.org/10.1080/17538157.2021.1904938
10. Mkungudza, Jonathan, Twabi, Halima S., Manda, Samuel O. M. (2024) Development of a diagnostic predictive model for determining child stunting in Malawi: a comparative analysis of variable selection approaches. *BMC Medical Research Methodology*. https://doi.org/10.1186/s12874-024-02283-6